Problem Solving through Programming (CSE103)
Multiple Choice Questions
Unit- 1

Q.No		
1`	Which of the following is a complete and functional setup that includes hardware, software, data, and users?	
	a) Monitor	b) Operating System
	c) Input Device	**d) Computer System**
2	Information provided by user to computer is called	
	a) Data	b) Output
	c) Compute	d) Result
3	A Computer program that converts assembly language to machine language is called	
	a) Compiler	b) Linker
	c) Assembler	d) Translator
4	The process of correcting errors in a program is called	
	a) Debugging	b) Testing
	c) Coding	d) Linking
5	Which of the following is an Input device?	
	a) Keyboard	b) Speaker
	c) Printer	d) CPU
6	Which step involves planning the program's structure?	
	a) Coding	b) Testing and Debugging
	c) Design (Algorithm)	d) Documentation
7	What is the purpose of 'Documenting the Program'?	
	a) Fixing errors in the program	b) Designing flowcharts
	c) Creating user and technical guides	d) Testing the final product
8	What is the final step of the program development cycle?	
	a) Design	**b) Maintenance**
	c) Debugging	d) Implementation
9	What is the main purpose of a programming language?	
	a) To play music	**b) To issue a sequence of commands to a computer**
	c) To display images	d) To write stories
10	Which of the following is considered a high-level programming language?	
	a) Assembly	b) Machine Code
	c) Python	d) Binary
11`	A Computer program that converts assembly language to machine language is called	
	a) Compiler	b) Linker
	c) Assembler	d)Translator
12	Step by step procedure to solve a give problem is called	
	a) Flowchart	**b) Algorithm**
	c) Machine program	d) Compiler

13	The ways that specific words and symbols are used by each language is called	
	a) Program	b) Algorithm
	c) Syntax	d) Grammar
14	What is #include <stdio.h>?	
	(a)Preprocessor directive	b) Inclusion directive
	c)File inclusion directive	d)None of the mentioned
15	Which escape character can be used to begin a new line in C?	
	a) \a	b)\m
	c) \n	d)\l
16	If a function in C does not return a value, what should be its return type?	
	a) Void	b) float
	c) int	d) null
17	The program written by the programmer in high level language is called	
	a) Object Program	**b) Source program**
	c) Compiled program	d) Assembled program
18	The format identifier '%x' is used for _____.	
	A) octal	B) decimal
	C) binary	**D) Hexadecimal**
19	Which of the following causes compilation error?	
	a) const int x = 10;	**b) int int = 5;**
	c) int a = sizeof('A');	d) float f = 1.0f + 2.0f;
20	Given the following line, how many **tokens** does it contain? int x = a + b * (c - d);	
	a) 11	b) 10
	c) 13	d) 8
21	What is a valid variable declaration?	
	a) 1total	**b) total**
	c) 1_total	d) int

22	What is the output of the following code?
	#include <stdio.h>
	
	int main() {
	int a = 5;
	printf("%d\n", a++);
	return 0;
	}	

	a) 11	**b) 5**
	c) 12	d) Undefined behavior
23	Which of the following is **not a keyword** in C?	
	a) static	b) unsigned
	c) typeof	d) goto
24	What is the output?	

	#include <stdio.h>
	int main()
	{
	int a, b=2;
	a= b << 2 + 3;
	printf("value of a is=%x\n", a);
	return 0;
	}	

	a) 2	b) 3
	c) 5	**d) 40**
25	The format identifier '%llu' is used for _____.	
	a) long	**b) long long unsigned**
	c) decimal	d) integer
26	Which function is used for formatted output in C?	
	a) printf()	b) scanf()
	c) getchar()	d) puts()
27	In C, scanf("%d", &x); reads	
	a) A floating-point number	**b) An integer**
	c) A string	d) A character
28	Which function is used for unformatted input of a single character?	
	a) scanf()	b) putchar()
	c) **getchar()**	d) printf()
29	Which of the following reads a string until a newline is encountered?	
	a) puts()	**b) gets()**
	c) getchar()	d) printf()
30	What does puts() do in C?	
	a) Reads a string from the user	**b) Writes a string to the console with a newline**
	c) Writes a string without a newline	d) Reads a single character
31	Which function is used for formatted input in C?	
	a) printf()	**b) scanf()**
	c) getchar()	d) puts()
32	The format specifier %s in scanf() is used for:	
	a) Single character	**b) String**
	c) Integer	d) Floating-point number
33	What will printf("%5d", 12); output?	
	a) 12	**b) 12 (3 spaces before 12)**
	c) 12 (3 spaces after 12)	d)Error
34	Which of the following is TRUE about scanf()?	
	a) It can take input without specifying format specifiers	**b) It stops reading a string at whitespace**
	c) It automatically trims extra spaces	d) It ignores newline characters

35	In command-line arguments, argv[0] contains:	
	a) First user argument	**b) Program name**
	c) Total number of arguments	d) Last user argument
36	Which of the following operators is used to perform a bitwise right shift?	
	a)<<	**b)>>**
	c)&	d)|
37	If x = 5, what will be the value of x << 2?	
	a)10	**b)20**
	c)15	d)25
	c)Compile time error	d)Run Time Error
38	What will be the data type of the following expression? (Initial data type: a = int, var1 = double, var2 = float) expression (a < 50)? var1 : var2;	
	a)int	b)float
	c)double	d)cannot be determined
39	Which of the following arithmetic operator takes only integer operands?	
	a)+	b)-
	c)/	**d)%**
40	Which of the following operators has the lowest priority?	
	a)&&	b)+
	c)*	d)!==
41	What will be the result of the following expression if a = 6, b = 3, and c = 4? (a * b - c) / (b + c % a) + (a % b * c)	
	a)7	b)8
	c)6	**d)10**
42	What will be the result of the following expression if x = 10 and y = 4? (x / y) == 2 && (x % y) > 0	
	a)True	b) False
	c)10	d)20
43	Which of the following operators is right-associative?	
	a)+ addition	b)* multiplication
	c)- subtraction	**d) ?: Ternary conditional**
44	Which of the following has the lowest precedence in C?	
	a) Logical AND (&&)	b) Assignment (=)
	c) Conditional (?:)	**d) Comma (,)**

45	Given:
	int a = 5;
	float b = 2.0;
	printf("%f", a / (int)b);
	What is printed?	

	a) 2.5	b) 2.0
	c) 2.000000	d) Compiler Error

46	What is the associativity of the conditional (?:) operator in C?	
	a) Left to right	**b) Right to left**
	c) No associativity	d) Same as logical operators
47	Which is true about implicit type conversion in C?	
	a) A float is promoted to double in arithmetic expressions	b) char is not converted to int automatically
	c) Conversion always happens from higher to lower precision	d) No conversion happens between signed and unsigned types

48	Evaluate:
	int x = 7;
	int y = 2;
	printf("%d", x % y * x / y);	

	a) 1	b) 0
	c) 2	**d) 3**
49	In C, the expression a = b = c = 5; assigns:	
	a) 5 to a only	b) 5 to a and b, but not c
	c) 5 to all a, b, and c	d) 5 to c only
50	Given int a = 5; float b = 2; what is the result of a / b?	
	a) 2	**b) 2.5**
	c) 2.0	d) 0